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By:

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Abstract

Introduction

2.1 Background and Motivation

The video game industry has experienced rapid growth over the past decade, with thousands of new titles released annually across multiple platforms. As the market expands, the pricing of video games has become increasingly complex, influenced by factors such as genre, release year, content availability, and overall popularity. Consumers often face difficulty determining whether a game is fairly priced, while developers and publishers must carefully balance pricing strategies to remain competitive and profitable.

With the availability of large-scale datasets such as SteamDB, it is now possible to analyze historical pricing data and uncover patterns that influence game pricing. Understanding these patterns can provide valuable insights into how different features contribute to price variation and can support more informed decision-making for both consumers and developers.

2.2 Problem Statement

Despite the abundance of available data, there is limited clarity on how specific game characteristics influence pricing. Factors such as genre, downloadable content (DLC), release year, and platform compatibility may all contribute to price differences, but their exact relationships are not always obvious.

The problem addressed in this project is to determine how various features of video games affect their pricing and to develop predictive models that can estimate a game's price or classify it into pricing categories. This involves both analyzing historical data and applying machine learning techniques to identify meaningful patterns and relationships.

2.3 Project Goals

The primary goals of this project are:

- To analyze how different features influence video game pricing
- To identify trends and patterns in pricing over time
- To build machine learning models that predict the price of a game
- To classify games into pricing categories (premium vs. budget)
- To gain insights into which factors are most significant in determining price

These goals combine exploratory data analysis with predictive modeling to better understand pricing behavior in the video game industry.

Dataset Description

3.1 Data Source

The dataset used in this project was obtained from SteamDB, a publicly available database that provides detailed information about games on the Steam platform. SteamDB includes a wide range of attributes related to video games, such as pricing information, release dates, genres, and additional content.

This dataset serves as a comprehensive source for analyzing pricing trends and relationships between game characteristics and price.

3.2 Features and Variables

The dataset contains a variety of features describing each game. Key variables include:

- **Final Price:** The current price of the game (used as the primary regression target)
- **Initial Price:** The original price before discounts

- **Release Year:** The year the game was released
- **Genre/Tags:** Categories that describe the type of game
- **DLC Count:** The number of downloadable content items associated with the game
- **Platform Information:** Compatibility with different systems
- **Other Numerical Features:** Metrics related to popularity or engagement

For the purposes of machine learning, features that directly reveal or strongly correlate with the target variable (such as final price, discount-related values, and free indicators) were removed to prevent data leakage.

3.3 Data Cleaning and Preprocessing

Before analysis and modeling, the dataset underwent several preprocessing steps to ensure quality and consistency. First, columns that could introduce data leakage were removed. These included variables directly related to pricing outcomes, such as final price, initial price, discount percentage, and discount amount. Removing these features ensures that the models do not rely on information that would not be available in a real prediction scenario. Next, only numerical and boolean features were retained for modeling. Boolean values were converted into integer format to ensure compatibility with machine learning algorithms.

Missing values in numerical features were handled using median imputation based on the training data. This approach provides a robust estimate that is less sensitive to outliers and helps maintain the integrity of the dataset. These preprocessing steps ensured that the dataset was clean, consistent, and suitable for both exploratory analysis and machine learning tasks.

Exploratory Data Analysis (EDA)

4.1 Overview of Data Distributions

Exploratory Data Analysis was conducted to better understand the structure and characteristics of the dataset. Initial analysis focused on the distribution of key variables, particularly game prices.

The distribution of game prices was found to be skewed, with a large number of lower-priced games and fewer high-priced titles. This indicates that most games fall within a lower price range, while premium games represent a smaller portion of the dataset.

Understanding these distributions is important for selecting appropriate modeling techniques and evaluation metrics.

4.2 Analysis of Pricing Trends

Further analysis examined how game prices vary across different factors such as release year, genre, and content availability.

Trends over time suggest that pricing may change based on release year, with newer games often priced higher due to increased development costs and demand. Additionally, games with more content, such as those with a higher number of DLCs, may tend to have higher prices.

These trends highlight the importance of considering multiple variables when analyzing pricing behavior.

4.3 Research Questions and Visualizations

To guide the analysis, several research questions were explored using visualizations:

1. **How do game price distributions vary across genres?**

   Visualizations such as boxplots were used to compare price distributions across different genres, revealing variations in pricing patterns.

2. **How have game prices changed over time based on release year?**

   Line plots and scatter plots were used to examine trends in pricing over time.

3. **Do games with more DLCs tend to have different base prices?**

   Scatter plots were used to explore the relationship between DLC count and price.

4. **Do higher development budgets lead to higher game prices?**

   Where available, relevant features were analyzed to assess this relationship.

5. **How are prices affected by development time?**

   Relationships between release timing and pricing were explored using appropriate visualizations.

These visualizations helped uncover relationships between features and pricing, providing a foundation for model development.

4.4 Key Insights from EDA

The exploratory analysis revealed several important insights:

- Game prices are highly skewed, with most games priced at lower ranges
- Pricing varies significantly across genres
- There is evidence that newer games tend to have higher prices
- Games with additional content, such as DLCs, may be associated with higher pricing
- Multiple factors interact to influence pricing, indicating that simple models may not fully capture these relationships

These findings support the need for machine learning models that can capture complex relationships between variables and accurately predict pricing outcomes.

Methodology

5.1 Data Preparation

The Steam dataset used in this project was first preprocessed to ensure it was suitable for machine learning tasks. Since the goal was to predict game prices and classify games into pricing categories, it was important to carefully select features that would not introduce data leakage. Columns directly related to pricing outcomes such as final price, initial price, discount percentage, discount amount, and indicators like whether a game is free were removed from the feature set. These variables either directly determine the target variable or are derived from it, and including them would lead to artificially inflated model performance.

Only numerical and boolean features were retained for modeling. Boolean values were converted into integer format to ensure compatibility across all machine learning models. Missing values in numerical columns were handled using median imputation based on the training data to avoid bias from extreme values.

5.2 Target Variable Construction

Two target variables were created to support both regression and classification tasks:

- Regression Target: The continuous variable representing the final price of a game.
- Classification Target: A binary variable indicating whether a game is considered “premium” or “budget.”

To construct the classification target, a threshold of $20 was used. Games with a final price greater than or equal to this threshold were labeled as “premium,” while those below were labeled as “budget.” This threshold was selected as a reasonable cutoff to distinguish between lower-cost and higher-cost games on the platform.

5.3 Train/Test Split

The dataset was divided into training and testing sets using an 80/20 split. The training set was used to train the machine learning models, while the testing set was reserved for evaluating final model performance on unseen data. To ensure consistency across both regression and classification tasks, a single split was applied to all features and target variables. Additionally, stratified sampling was used based on the classification target to preserve the proportion of premium and budget games in both the training and testing sets. This helps prevent class imbalance from affecting model evaluation.

A fixed random seed was used during the split process to ensure reproducibility of results.

5.4 Feature Engineering and Pipeline

Feature preprocessing was designed to standardize the data and prepare it for modeling. Numerical features were scaled where necessary, and categorical variables were transformed into numerical representations using encoding techniques. To maintain consistency across models, preprocessing steps were designed to be reusable and applied uniformly to both training and testing data. This ensures that all models are trained and evaluated under the same conditions. In addition, we wanted to avoid data leakage by ensuring that preprocessing steps such as imputation were fitted only on the training data and then applied to the test data.

5.5 Model Training Strategy

Multiple machine learning models were trained to address both regression and classification objectives. For regression, models were used to predict the price of a game, while classification models were used to determine whether a game falls into the premium or budget category.

To evaluate model performance, **k-fold cross-validation** was applied to the training data. This approach involves dividing the training data into multiple folds and training the model on different subsets while

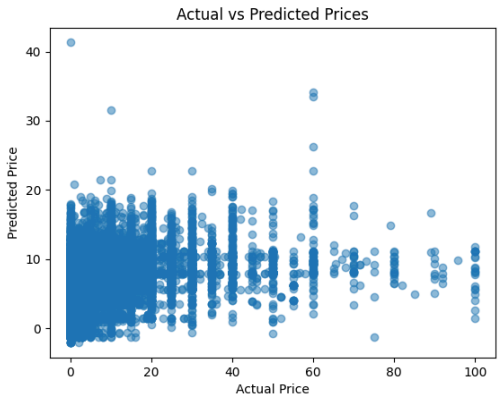
validating on the remaining data. This provides a more reliable estimate of model performance compared to a single train/test split. All models were trained using the same training data and evaluated using consistent metrics to allow for fair comparison.

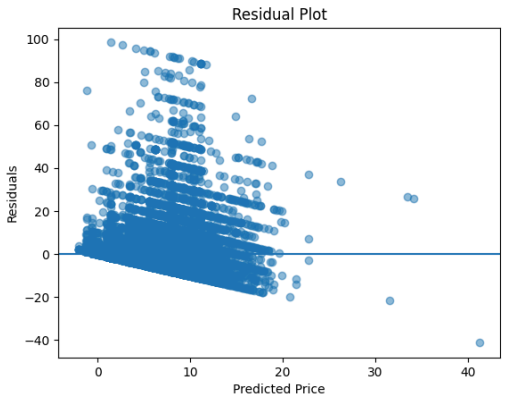
5.6 Evaluation Approach

After training and cross-validation, the best-performing models were selected based on their evaluation metrics. These models were then tested on the held-out test set to assess their performance on unseen data. For regression models, performance was evaluated using metrics such as Mean Squared Error (MSE) and R^2 score. For classification models, evaluation metrics included accuracy, precision, recall, and F1-score. A confusion matrix was also used to provide a detailed view of classification performance. This structured evaluation approach ensures that model performance is both reliable and generalizable.

Machine Learning Models

6.1 Linear Regression





Linear regression was implemented as a baseline model to predict game prices using numerical and encoded features. Both scaled and unscaled versions were tested, and 5-fold cross-validation was applied to evaluate model stability. Scaling did not significantly impact performance, which is expected for linear regression.

6.2 Random Forest

6.3 Logistic Regression

Model Evaluation

7.1 Evaluation Metrics (Regression)

Linear Regression: Model performance was evaluated using Mean Squared Error (MSE) and R^2. The model achieved an R^2 of approximately 0.14 and an MSE of about 70, indicating a limited ability for prediction. 5-fold cross-validation produced an average R^2 of approximately 0.13, with consistent results across folds. This suggests that the model's performance is stable but consistently low, indicating poor generalization. Feature scaling did not significantly impact performance since both scaled and unscaled models produced nearly identical results.

7.2 Evaluation Metrics (Classification)

7.3 Model Comparison

7.4 Best Model Selection

Results

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